NBA GamePlan:
An Interactive Matchup Predictor & Strategy Analyzer

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Motivation:

Modern sports analytics has grown into a major part of how teams, analysts, and fans understand performance. Most existing models focus only on accuracy and provide little insight into *why* certain outcomes occur. This lack of transparency limits their usefulness for people who want to understand what actually drives success.

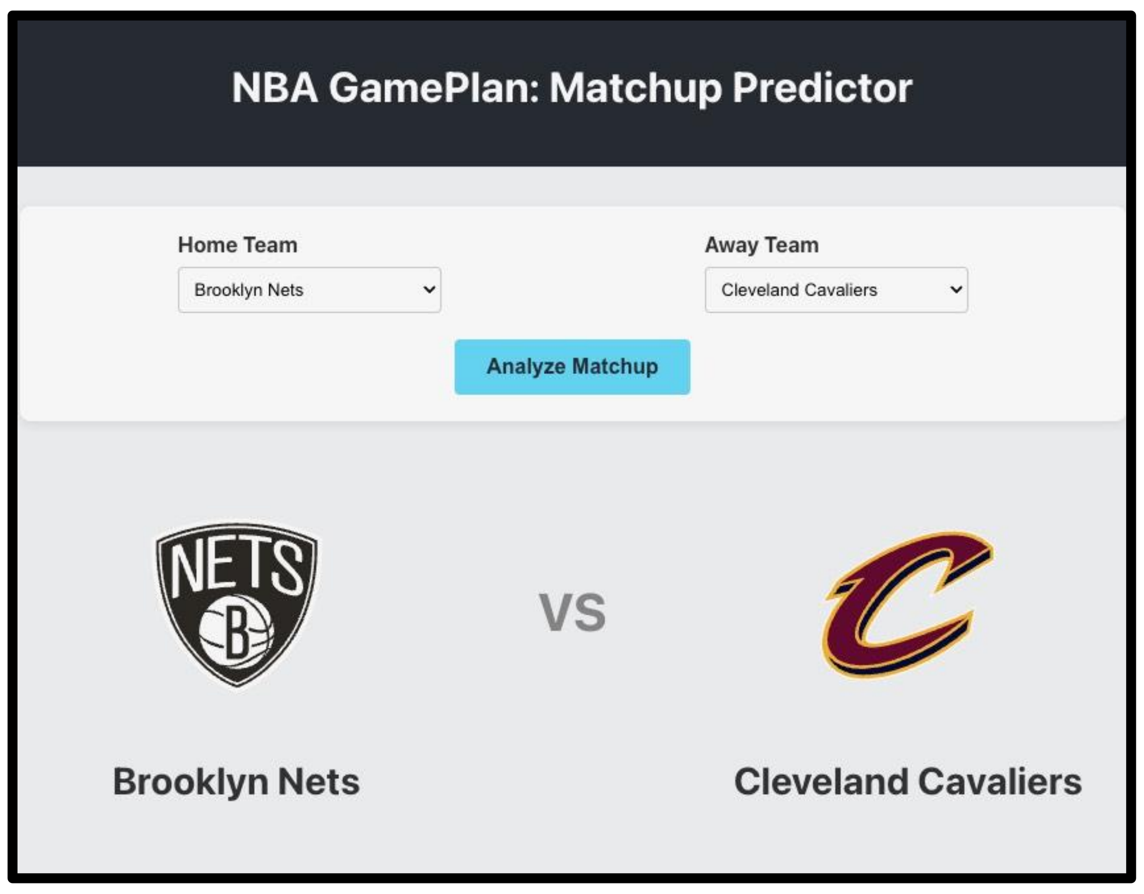

Figure 1: Team selection user interface

Methodology:

1. Focus on 2020–2025 NBA data using exponentially weighted moving averages (EWMA) with span=10 for team stats and span=5 for player stats to emphasize recent performance
2. Train XGBoost classification and regression models, both with 300 estimators, max depth 7, and learning rate 0.05
3. Initialize SHAP explainers for interpretability.
4. Build a dashboard with team selection, prediction results, season stats, and SHAP feature explanations.

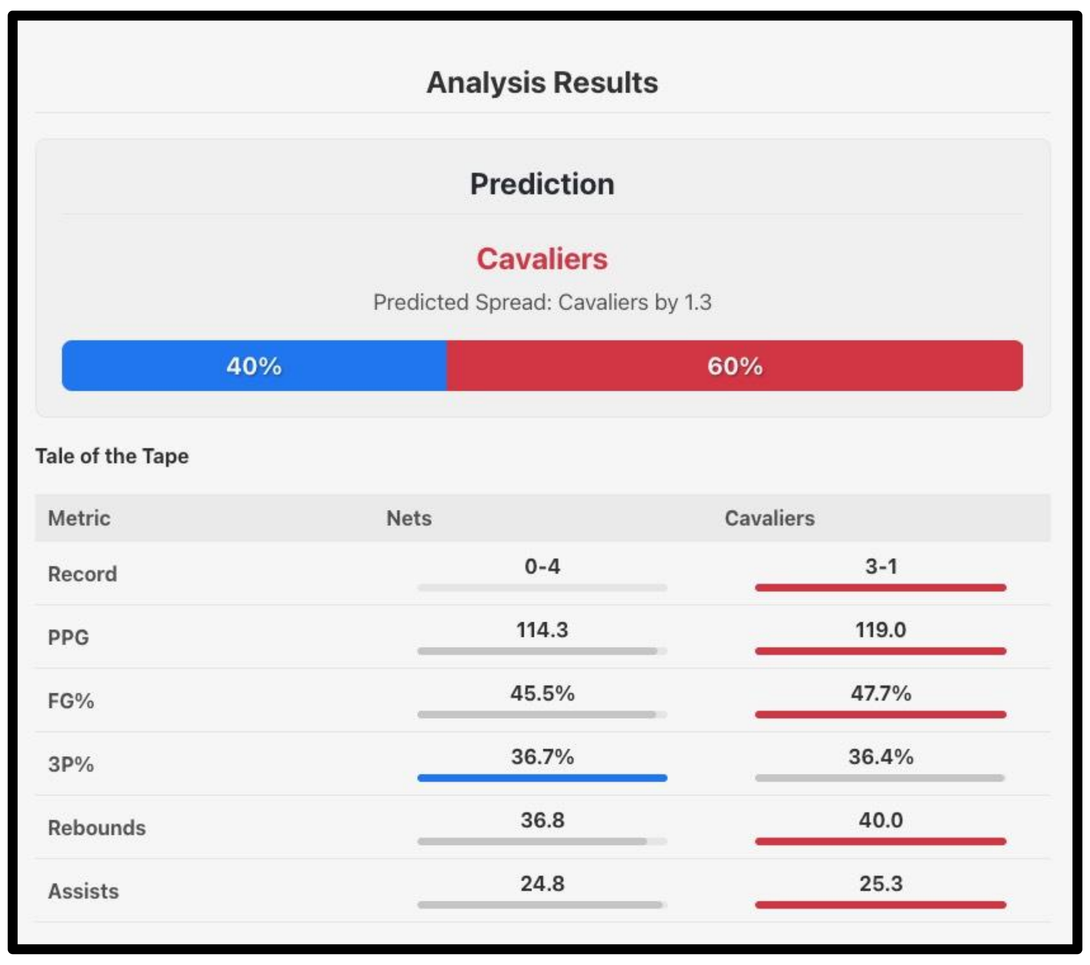

Figure 2: Prediction Analysis

Approach:

We use a combination XGBoost with SHapley Additive exPlanations(SHAP), to supply in depth explanations as well as predictions. We combine player and team statistics, while using Exponentially Weighted Moving Averages(EWMA) to provide emphasis on recent performance. Coupled with a easily navigable UI and accompanying data visualizations, this provides a beginner friendly way to analyze games, with enough depth for seasoned analyst as well.

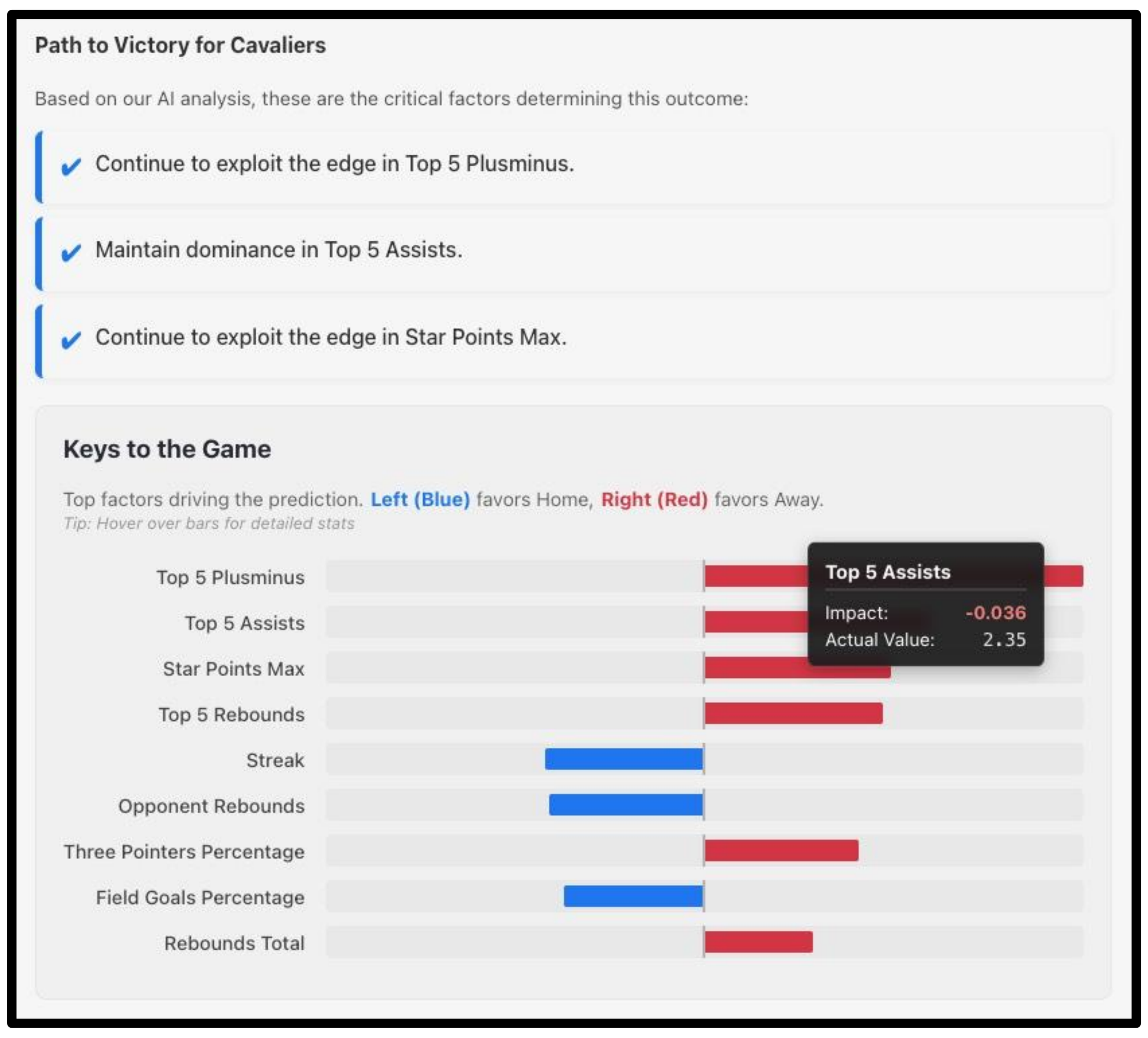

Figure 3: SHAP Results

Our Data:

NBA Dataset - Box Scores & Stats, 1947 - Today

Updated daily. Player & team statistics since 1947. Current / all past games.

kaggle

Kaggle - Historical NBA data and player box scores

Temporal dataset downloaded locally; contains 50+ years worth of team and player statistics

Features include home, teamScore, points, assists, etc.

Results:

- Accuracy of roughly 72 %
- Precision ≈ 0.71
- Recall ≈ 0.70
- AUC of 0.78 on the validation split
- MAE of 5.1 points
- RMSE of 6.8 points for spread prediction.
- Compared to baselines, logistic regression reached ≈ 64 % accuracy and random forest ≈ 68 %

User Case Study Findings

Moderate-to-low basketball fans liked the clear color-coding and the simplicity of the “Path to Victory” section

Basketball enthusiasts valued the “Keys to the Game” section because it provided deeper insights into model’s decision making, along with ranking and magnitude of each feature.